COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 22 — PERT / CPM, Critical Path & Schedule Estimation

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Section 10: Classroom Activity — Calculate a Critical Path

Step-by-step Critical Path Method (CPM) calculation for the CEMS development network (Activities A to G, all finish-to-start dependencies).

1. Project Activity Network Data

Activity Code	Activity Description	Duration (Days)	Predecessor Activities
A	Requirements Engineering	2	None
B	User Interface Design	3	A
C	Database Schema Design	2	A
D	Build User Interface	4	B
E	Build Backend APIs	5	C
F	System Integration	2	D, E
G	System & Acceptance Testing	3	F

2. Forward Pass & Backward Pass Calculations

Activity	Duration	ES	EF	LS	LF	Total Float (LS - ES)	Critical Path?
A	2	0	2	0	2	0	YES (Critical)
B	3	2	5	4	7	2	NO
C	2	2	4	2	4	0	YES (Critical)
D	4	5	9	7	11	2	NO
E	5	4	9	4	9	0	YES (Critical)
F	2	9	11	9	11	0	YES (Critical)
G	3	11	14	11	14	0	YES (Critical)

3. Critical Path Identification & Delay Analysis

Critical Path: Path A -> C -> E -> F -> G. Total Expected Duration: 2 + 2 + 5 + 2 + 3 = 14 Days. Path A -> B -> D -> F -> G has duration 2 + 3 + 4 + 2 + 3 = 14 Days (Dual critical path under exact durations; Activity C-E has 0 float).
Risk Analysis: Any 1-day delay in Activity E (Build APIs) or Activity C (Database Design) directly delays overall project completion by 1 day because their Total Float is strictly 0.

Section 13: Homework — CPM & PERT Calculations for Library System

Probabilistic PERT analysis and CPM schedule estimation for an 8-activity Library Management System.

1. PERT Three-Point Duration & Variance Calculations

Formula: Expected Duration $T_e = \frac{O + 4M + P}{6}$; Variance $\sigma^2 = \left(\frac{P - O}{6}\right)^2$

Activity	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Expected Duration (Te)	Variance (Var)
Act 1: Database Setup	2 days	4 days	6 days	4.00 days	0.44
Act 2: Core Circulation API	4 days	7 days	16 days	8.00 days	4.00
Act 3: Integration Testing	3 days	5 days	13 days	6.00 days	2.78

2. Schedule Slippage Response Strategy

If a critical-path activity slips by 2 days:
1. Fast-Tracking: Overlap downstream critical-path activities that were planned sequentially (e.g., commence integration test preparation concurrently during the final 2 days of API construction).
2. Crashing: Assign a senior backend engineer to assist on the stalled critical activity (allocating developer overtime) after verifying that communication overhead will not violate Brooks’s Law.